

22 C

Effective resistance of the 3 parallel resistors,

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DUNMAN HIGH SCHOOL
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$$R_{\text{eff}} = \left(\frac{1}{1.0} + \frac{1}{2.0} + \frac{1}{5.0} \right)^{-1} = 0.588 \, \Omega$$

Potential difference across each resistor, $V = IR_{\text{eff}} = (5.0)(0.588) = 2.94 \, \text{V}$

Current through $2.0 \, \Omega$ resistor, $I_2 = V/R = 2.94/2.0 = 1.47 \approx 1.5 \, \text{A}$